

PARAMETER ESTIMATION TECHNIQUES

Estimating parameters of the I-Star model is not as direct as estimating the betas of a linear regression model. The I-Star model is non-linear and dependent on the product of parameters and these parameters are extremely sensitive in the marketplace. As a result it is difficult to find robust parameters that are stable over time.

For example, our model is:

$$I^* = a_1 \cdot \left(\frac{Q}{ADV} \right)^{a_2} \cdot \sigma^{a_3}$$

$$MI = b_1 \cdot I^* \cdot POV^{a_4} + (1 - b_1) \cdot I^*$$

After substitution we have:

$$MI = b_1 \cdot \left(a_1 \cdot \left(\frac{Q}{ADV} \right)^{a_2} \cdot \sigma^{a_3} \right) \cdot POV^{a_4} + (1 - b_1) \cdot \left(a_1 \cdot \left(\frac{Q}{ADV} \right)^{a_2} \cdot \sigma^{a_3} \right)$$

Below are three different techniques that have been used in the industry to estimate the parameters of this model. These techniques are:

1. Two-Step Process
2. Guesstimate Technique
3. Non-Linear Least Squares

Technique 1: Two-Step Process

The two step regression process involves fitting the parameters of the models in two stages. This technique was developed for the case where $a_4 = 1$ and there is a linear relationship between temporary impact and trading cost. This is as follows:

Step 1: Estimate Temporary Impact Parameter

Group the data into different POV rate categories such as 1% buckets. It is important to have a balanced data set so that the sizes and volatilities in each bucket are consistent across different POV rates. Otherwise, sample the data such that each POV bucket has identical order size and volatility characteristics.

One method to fit parameters is to group data into size and volatility buckets and compute each category's average market impact. This grouping is useful